Red Pitaya programming for van Exter group

Siglent SSG3032X (signal generator)

This produces an RF signal we want to modulate. The modulation should be synced with the other components

There are 3 ways to externally control the modulation

1) External amplitude modulation input

Needs a signal from -1V to 1V (2Vpp) for modulation, in which 1V is max amplitude (on), and -1V is zero amplitude (off).

Downside : The imperfections of the modulation signal (things like overshoot) get copied to the RF signal

2) Pulse in

Triggers around 1.6V (CMOS 3.3 V).

Expects the full pulse train, with low being off, and high being on (The Siglent can also be configured to inverse this).

Pulse modulation		
Modulation source	internal, external	
On/off ration	1 MHz <f<3.2 GHz	>70 dBc
Raise/fall time (10% / 90%)	10% to 90% of RF amplitude	<50 ns
Pulse repetition time	setting range	40 ns~300 s
Pulse generator		
Pulse modes	single pulse, double pulse	
Pulse source	internal, external	
Pulse polarity	normal, inverse	
Pulse period	setting range	40 ns~300 s
	resolution of setting	10 ns
Pulse width	setting range	20 ns~300 s
	resolution of setting	10 ns
Double pulse Delay	setting range	20 ns~300 s
	resolution of setting	10 ns
#2 Width	setting range	20 ns~300 s
	resolution of setting	10 ns
Trigger modes	auto, keyboard, external trigger, external gate trigger, bus	
Trig polarity	normal, inverse (used in external gate trigger mode)	
Trigger Slop	positive, negative (used in external trigger mode)	
External trigger delay	140 ns~300 s	
External trigger delay resolution of setting	10 ns	

3) Trigger in

Triggers around 1.6V (5 V TTL)

Expects a trigger pulse (can be configured to trigger on rising or falling edge) when the pulse train needs to start. A delay can be configured.

Downside: The pulse train would need to be programmed into the Siglent.

Our preference is option 2, where the pulse train is programmed into the Red Pitaya and then delivered to the Siglent as on/off commands.

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Timeharp 260 P (photon counter)

This detects/counts the pulses made by the photon detectors. This needs to get a sync pulse so it knows when the measurement starts.

The sync pulse is currently controlled by the pulse out of the Siglent (CMOS 3.3 V), but this needs to now be done by the Red Pitaya.

The two (= signal and sync) inputs of the Timeharp both trigger on the edge of a signal between 0 and -1 V (at trigger levels that can be adjusted in the Timeharp program). Hence, we typically use an invertor and a set of attenuators (in total 10 dB or more) to covert the TTL (0 to +5 V) signals from Siglent and photon counter to the required input.

Acousto-optical modulator (AOM) (laser modulation)

Expects a TTL signal with high turning the deflection of the laser on, and low turning the deflected laser beam off. (Note: we don't have the AOM in the lab yet and the on and off signals might be the other way around)

Alternatively, it can be provided with an analog signal between 0 and 5V to manually provide the modulation signal.

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Signals to be generated

1) A sync pulse for the Timeharp, once per measurement.

Preferably timed simultaneously with the read-pulse for the AOM, but the most important part is that it is consistent.

2) A configurable pulse train for the Siglent

A pulse train of 1 or more pulses. The pulse widths will be either t_π or t_π /2, with t_π being order 0.1 -1 µs long.

The distance between pulses will be τ, order 1 - 30 µs long.

A typical pulse sequence will look like this: (i) $t_\pi/2$ pulse, (ii) wait time τ, (iii) t_π pulse, (iv) wait time τ, (v) $t_\pi/2$ pulse (see figure below). Alternative pulse sequence start from this sequence but contain (N-1) additional copies of the block (ii)+(iii)+(iv), such that the N intermediate t_π pulses will be separated from each other by a time 2 τ.

It would be nice to also be able to do a simpler scheme with a t_{on} an t_{off} (basically one long pulse), both order 1-100 µs long.

3) A configurable pulse train for the AOM

The AOM needs 1 or 2 pulses per measurement: a prepare and read pulse, or a combined read-prepare pulse. These need to be order microseconds long. It would be nice to be able to put the read pulse as close as possible after the last pulse send to the Siglent.

It would be nice to also be able to forgo the modulation entirely.

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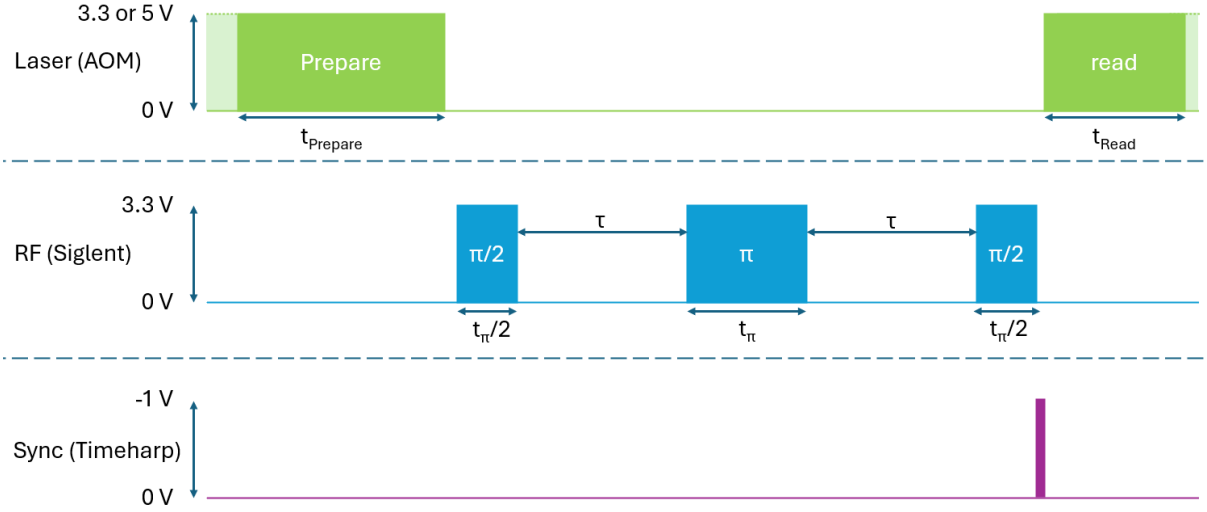
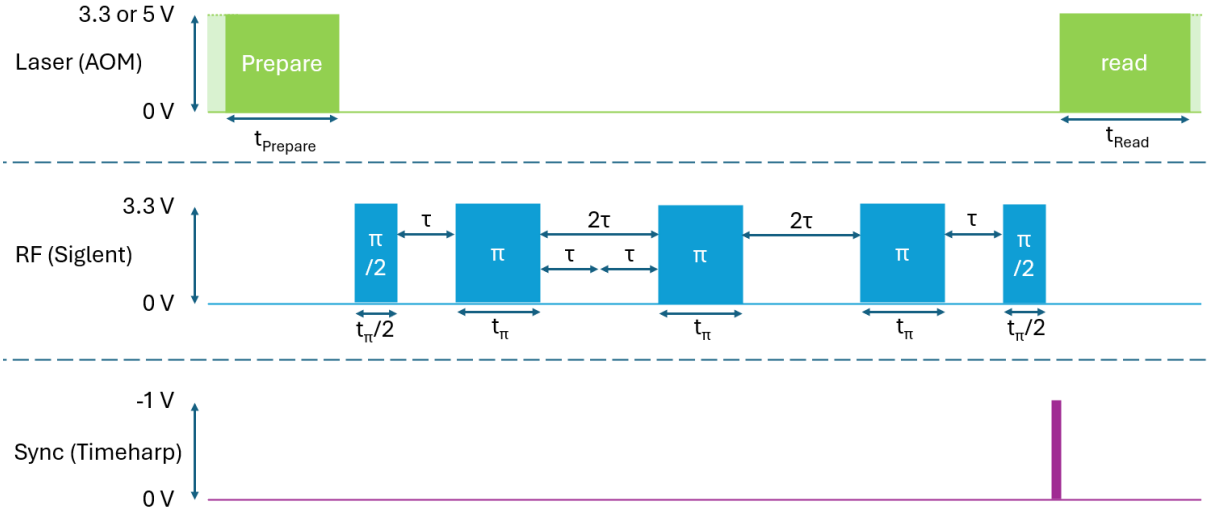
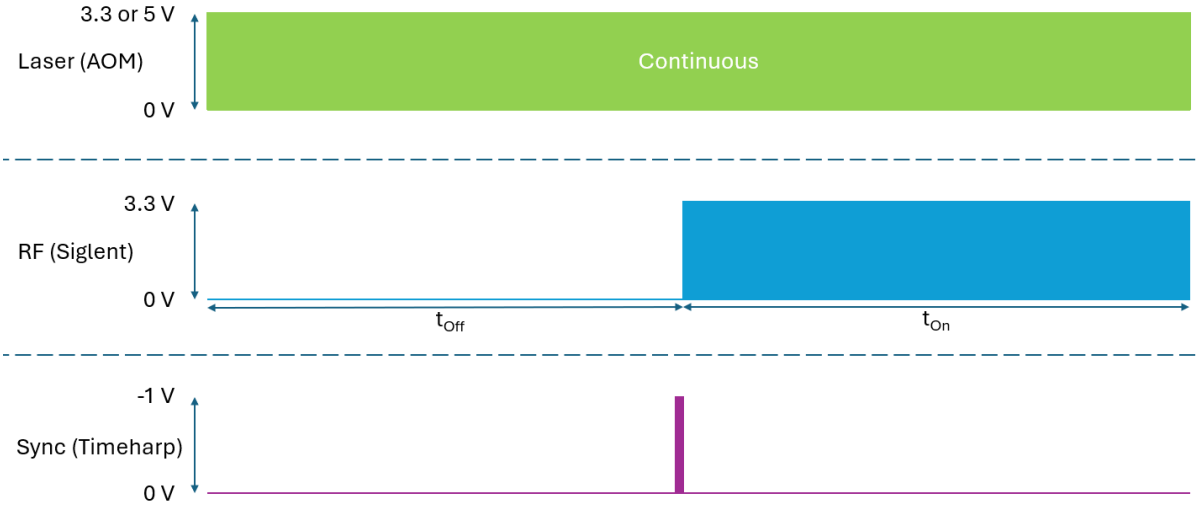


Example of an experiment (Hahn echo) we want to do with this setup.



Example of an experiment (N=3 Repeated pulses) we want to do with this setup.



Example of an experiment (Rabi Oscillation) we want to do with this setup.

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Rear panel connectors		
TRIG IN / OUT	impedance	100 kΩ
	connector	BNC female
	active trigger voltage	5 V TTL
EXT MOD INPUT	impedance	50 Ω
	connector	BNC female
PULSE IN / OUT	impedance	100 kΩ
	connector	BNC
	input/output voltage	CMOS 3.3 V

Taken from the Siglent datasheet